

Radar automatic target recognition based on feature extraction for complex HRRP

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Radar high-resolution range profile (HRRP) has received intensive attention from the radar automatic target recognition (RATR) community. Usually, since the initial phase of a complex HRRP is strongly sensitive to target position variation, which is referred to as the initial phase sensitivity in this paper, only the amplitude information in the complex HRRP, called the real HRRP in this paper, is used for RATR, whereas the phase information is discarded. However, the remaining phase information except for initial phases in the complex HRRP also contains valuable target discriminant information. This paper proposes a novel feature extraction method for the complex HRRP. The extracted complex feature vector, referred to as the complex feature vector with difference phases, contains the difference phase information between range cells but no initial phase information in the complex HRRP. According to the scattering center model, the physical mechanism of the proposed complex feature vector is similar to that of the real HRRP, except for reserving some phase information independent of the initial phase in the complex HRRP. The recognition algorithms, frame-template establishment methods and preprocessing methods used in the real HRRP-based RATR can also be applied to the proposed complex feature vector-based RATR. Moreover, the components in the complex feature vector with difference phases approximate to follow Gaussian distribution, which make it simple to perform the statistical recognition by such complex feature vector. The recognition experiments based on measured data show that the proposed complex feature vector can obtain better recognition performance than the real HRRP if only the cell interval parameters are properly selected.

complex high-resolution range profile (HRRP), radar automatic target recognition (RATR), feature extraction, minimum Euclidean distance classifier, adaptive Gaussian classifier (AGC)

1 Introduction

A high-resolution range profile (HRRP) is the coherent summations of the complex time returns

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from target scatterers in each range cell, which represents the projection of the complex returned echoes from the target scattering centers onto the radar line-of-sight (LOS). It contains target structure signatures, such as target size, scatterer distribution, etc., thereby radar HRRP target recognition has received intensive attention from the radar automatic target recognition (RATR) community^[1–10]. Nevertheless, according to the definition of HRRP given in refs. [1–10], an HRRP is a real vector, which is the amplitude of complex returned echo vector in baseband. Obviously, these literatures did not apply the phase information in the complex returned echo to RATR.

Assume that a point target's returned echo in radio frequency is $s(t-\tau)e^{j\omega_c(t-\tau)}$, where $s(\cdot)$ is the complex envelope, ω_c is the carrier angular frequency, $\tau = 2R/c$ is the time delay with c denoting the speed of light and R representing the radial distance between the target and radar, thereby the returned echo in baseband is $s(t-\tau)e^{-j\omega_c\tau}$, where the phase information includes two parts, i.e., the phase of the complex envelope $s(\cdot)$, which depends on the target scattering property and can be useful for RATR; and the initial phase, i.e., $\psi_\tau = -\omega_c\tau = -(4\pi/\lambda)R$ with λ denoting the radar wavelength, which depends on the target distance and the radar wavelength and does not contain the target discrimination information. Furthermore, the initial phase is strongly sensitive to the target position variation. For example, for carrier angular frequency of $2\pi \times 5520$ MHz (approximately, the wavelength of 5 cm), if the radial displacement of a target is 2.5 cm, the initial phase difference will be 2π . In applications, the measurement precision of the target distance usually cannot satisfy the requirement of the initial phase compensation, especially for some complicated targets; therefore, the measured complex echo has the initial phase sensitivity. This is the analysis of the property of the phase information contained in a point target's complex echo. For high resolution radar system, since the targets and their components are much larger than the wavelength of microwave radar, a target does not appear as a point target any longer, but consist of many scatterers, each as a point target, separated in several range cells along the radar LOS, and the corresponding returned echo is the coherent summations of the complex time returns from target scatterers in each range cell, which is the scattering center model assumption^[11]. According to this model theory^[11,12], the phase information contained in the high-resolution complex echo from a target, namely the complex HRRP defined in the last paragraph, also includes two parts, i.e., the initial phase which is sensitive to the target position variation, and the remained phase which can reflect the target scattering property (detailed derivation shown in section 2. Therefore similar to the scatterer distribution information contained in the real HRRP, the phase information contained in the complex HRRP should also be valuable for RATR. However, since the initial phase is very sensitive to the variation of target's radial distance, it is hard to apply the target's phase information to RATR. Some literatures used simulated turntable HRRP data, e.g., ref. [6]. Since the targets are assumed to only rotate around the turntable center in the simulation, the literatures using simulated turntable HRRP data did not deal with the initial phase sensitivity problem. However, in applications, at least the test HRRP samples are measured during the target's movement. Until now, there has been little work in the field of the complex HRRP-based RATR. Ref. [6] directly applied some recognition algorithms for real HRRP-based RATR to the complex HRRP-based RATR under the condition of using simulated turntable data, yet without the special study on characteristics of the complex HRRP, and ref. [12]

proposed a special recognition method for the complex HRRP, which used the principal component analysis (PCA)-based minimum reconstruction error approximation to deal with the initial phase sensitivity of the complex HRRP, yet without study on how to apply other recognition algorithms used in the real HRRP-based RATR to the complex HRRP-based RATR.

Since a complex HRRP consists of the discrete echo components from different range cells, if the complex echo from one range cell is multiplied by the conjugate complex echo from another range cell, the initial phase will be counteracted and the corresponding difference phase, which is independent of the target position, will remain. By this property, a complex HRRP with some range cells' delay multiplied by the conjugate vector of the original complex HRRP will form a new complex feature vector with the difference phase information in the complex HRRP. Moreover, considering the influence of amplitude property on the recognition, and to simplify the statistical recognition, each product term, i.e., each components in the above conjugate product vector, should be divided by a corresponding conversion factor. Such method is referred to as a novel feature extraction method using the difference phase information in the complex HRRP, and the extracted complex feature vector is referred to as a complex feature vector with difference phases. Based on the scattering center model, the physical mechanism of the complex feature vector is similar to that of the real HRRP, except for containing the phase information. Therefore, the recognition algorithms, frame-template establishment methods and preprocessing methods used in the real HRRP-based RATR can also be applied to the proposed complex feature vector-based RATR. Moreover, the components of the complex feature vector with difference phases approximate to follow Gaussian distribution; therefore, the statistical recognition by such complex feature vector is simple. The recognition experiments based on measured data show that the proposed complex feature vector can obtain better recognition results than the real HRRP in the minimum Euclidean distance classifier and the adaptive Gaussian classifier (AGC) if only the cell interval parameters are proper.

2 Characteristic comparison between real HRRP and complex HRRP

Suppose there are radial displacement and little uniform rotation for a target. If the transmitted signal is $s(t)e^{j\omega_c t}$ and the returned complex HRRP is denoted as a discrete complex vector, the m th complex returned echo from the n th range cell ($n = 1, 2, \dots, N$) in baseband is

$$x_n(t, m) = \sum_{i=1}^{L_n} \sigma_{ni} s\left(t - \frac{2R_{ni}(m)}{c}\right) e^{-j\left[\frac{4\pi}{\lambda}R_m(m) - \psi_{ni0}\right]}, \quad (1)$$

where L_n denotes the number of target scatterers in the n th range cell, σ_{ni} and ψ_{ni0} represent the strength and the initial phase of the i th scatterer in the n th range cell, and $R_{ni}(m)$ denotes the radial distance between the radar and the i th scatterer in the n th range cell in the m th returned echo. If $R(m)$ denotes the radial distance between the target reference center in the m th returned echo and the radar, and L_y denotes the radial length of the target, usually $R(m) \gg L_y$. Due to the target rotation, there are radial displacements for the scatterers. If $\Delta r_{ni}(m)$ represents the radial displacement of the i th scatterer in the n th range cell in the m th returned echo, then $R_{ni}(m) \approx R(m) + \Delta r_{ni}(m)$. Because the complex envelope $s(\cdot)$ in a range cell approxi-

mates to be unvaried with the radial displacements for all scatterers in the range resolution cell, eq. (1) can be approximated as

$$\begin{aligned} x_n(t, m) &\approx x_n(m) = \sum_{i=1}^{L_n} \sigma_{ni} \exp \left\{ -j \left\{ \frac{4\pi}{\lambda} [R(m) + \Delta r_{ni}(m)] - \psi_{ni0} \right\} \right\} \\ &= e^{-j \frac{4\pi}{\lambda} R(m)} \sum_{i=1}^{L_n} \sigma_{ni} \exp \left\{ -j \left[\frac{4\pi}{\lambda} \Delta r_{ni}(m) - \psi_{ni0} \right] \right\} = e^{j\psi(m)} \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)}, \end{aligned} \quad (2)$$

where $\psi(m) = -\frac{4\pi}{\lambda} R(m)$ denotes the initial phase of the m th returned echo, which does not

contain the target discrimination information, and $\phi_{ni}(m) = \psi_{ni0} - \frac{4\pi}{\lambda} \Delta r_{ni}(m)$ denotes the remained echo phase of the i th scatterer in the n th range cell in the m th returned echo, which contains the scatterers' radial displacement information and can reflect the target's scatterer distribution property.

2.1 Characteristics of real HRRP

As the aforementioned derivation, the m th real HRRP can be defined as

$$\begin{aligned} x(m) &\triangleq [|x_1(m)|, |x_2(m)|, \dots, |x_n(m)|, \dots, |x_N(m)|]^T \\ &= \left[\left| \sum_{i=1}^{L_1} \sigma_{1i} e^{j\phi_{1i}(m)} \right|, \left| \sum_{i=1}^{L_2} \sigma_{2i} e^{j\phi_{2i}(m)} \right|, \dots, \left| \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)} \right|, \dots, \left| \sum_{i=1}^{L_N} \sigma_{Ni} e^{j\phi_{Ni}(m)} \right| \right]^T, \end{aligned} \quad (3)$$

where T denotes the transpose operation. Obviously, the real HRRP is unvaried with the initial phase, but some of the scatterer distribution information contained in the scatterers' echo phases is discarded by the modulus operation. As stated in refs. [7,8], it is a prerequisite of the real HRRP-based RATR to deal with the target-aspect, time-shift, and amplitude-scale sensitivity problems of the real HRRP. Refs. [7,8] gave the detailed analysis on and solution approaches to the three sensitivity problems.

Another important characteristic of the real HRRP is that the average profile of a real HRRP set from a target-aspect sector without scatterers' motion through range cells (MTRC), which is referred to as a real HRRP frame in this paper, can reduce the target-aspect sensitivity of the real HRRP samples in a large extent. According to eq. (2), the power of $x_n(m)$ can be written as

$$|x_n(m)|^2 = x_n(m) \cdot x_n^*(m) = \sum_{i=1}^{L_n} \sigma_{ni}^2 + 2 \sum_{i=2}^{L_n} \sum_{k=1}^{i-1} \sigma_{ni} \sigma_{nk} \cos[\phi_{ni}(m) - \phi_{nk}(m)] = Y_n + \xi_n(m), \quad (4)$$

where $*$ denotes the complex conjugation operation. In eq. (4), the first term Y_n is referred to as the scatterer auto-term (SAT), which is the summation of the scatterers' energy within a range cell; the second term $\xi_n(m)$ is referred to as the scatterer cross-term (SCT), which is the summation of the conjugate multiplication of echoes from different scatterers within a range cell. Without scatterers' MTRC, the SAT Y_n remains invariant, but the SCT $\xi_n(m)$ varies with m as a random variable with zero mean^[7,8], from which the target-aspect sensitivity of HRRP results. Therefore, in order to reduce the target-aspect sensitivity of real HRRP samples, the objective is in fact the reduction of the SCT. Let $\{x(0), x(1), \dots, x(m), \dots, x(M-1)\}$ denote a real HRRP frame. In applications, the average profile of a real HRRP frame is defined as

$$\bar{x}^{\Delta} = \left[\sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_1(m)|^2}, \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_2(m)|^2}, \dots, \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_n(m)|^2}, \dots, \sqrt{\frac{1}{M} \sum_{m=0}^{M-1} |x_N(m)|^2} \right]^T. \quad (5)$$

Now we show that the average profile can reduce the SCT in a real HRRP frame to a large degree from the ISAR image. Let $\tilde{x}(m) = [x_1(m), x_2(m), \dots, x_n(m), \dots, x_N(m)]^T$ represent the complex HRRP of the m th time return. Performing the Fourier transformation to each row of the motion compensated complex HRRP frame $\tilde{x} = [\tilde{x}(0), \tilde{x}(1), \dots, \tilde{x}(m), \dots, \tilde{x}(M-1)]$, which corresponds to each range cell, one can obtain the ISAR image $X = [X(0), X(1), \dots, X(k), \dots, X(M-1)]$, where $X(k) = [X_1(k), X_2(k), \dots, X_n(k), \dots, X_N(k)]^T$ and $X_n(k)$ represents the complex strength summation of the scatterers in the resolution gridding (n, k) , which is referred to as an image pixel. According to the energy preservation property of the Fourier transformation, i.e., the Parseval theorem

$$\sum_{k=0}^{M-1} |X_n(k)|^2 = \sum_{m=0}^{M-1} |x_n(m)|^2, \quad (6)$$

where $\sum_{m=0}^{M-1} |x_n(m)|^2$ is the summation of all power echoes in the n th range cell, the 0.5-powered term of which is identical with the n th component in the average profile after being amplitude normalized, and $\sum_{k=0}^{M-1} |X_n(k)|^2$ is the energy summation of the image pixels within the n th range cell, which is also equal to the stable SAT in eq. (4). Figure 1 is an example from measured “Yark-42” data. Figures 1(a) and 1(c) compare the 0.5-powered projection of the ISAR image onto the radial range dimensional and the 0.5-powered summation of all power profiles, which show that the two profiles are identical. The average profile defined in eq. (5) can represent the stable SAT approximately and reduce the SCT in the HRRP frame to a large degree. Therefore, the average real profiles can be used as the target-aspect templates in the real HRRP-based RATR.

2.2 Characteristics of complex HRRP

As the above derivation, the m th complex HRRP is

$$\begin{aligned} \tilde{x}(m) &= [x_1(m), x_2(m), \dots, x_n(m), \dots, x_N(m)]^T \\ &= e^{j\psi(m)} \left[\sum_{i=1}^{L_1} \sigma_{1i} e^{j\phi_{1i}(m)}, \sum_{i=1}^{L_2} \sigma_{2i} e^{j\phi_{2i}(m)}, \dots, \sum_{i=1}^{L_n} \sigma_{ni} e^{j\phi_{ni}(m)}, \dots, \sum_{i=1}^{L_N} \sigma_{Ni} e^{j\phi_{Ni}(m)} \right]^T. \end{aligned} \quad (7)$$

Obviously, being different from the real HRRP, the complex HRRP contains all of the information in the scatterers’ echo phases, including the initial phase and the scatterers’ remained echo phases. Therefore, besides the target-aspect, time-shift, and amplitude-scale sensitivities, as discussed in section 1, there is another sensitivity problem for complex HRRP, i.e., the initial phase sensitivity. Similar to the time-shift sensitivity, there are also two kinds of initial phase sensitivity in RATR using the complex HRRP. a) In the training phase, the initial phases of each training frame should be calibrated, which can be achieved by using the autofocus techniques in ISAR imaging, such as dominant scatterer algorithm (DSA) or its modifications^[11]. b) In the test phase, before a test sample is matched with a training frame, its initial phase should be calibrated with

the training samples' initial phases. Since the test sample and the training samples are not continuously measured, the autofocus techniques in ISAR imaging are not suitable for this process. Similar to the time-shift compensation in the test phase, the optimal initial phase compensation of the test sample should be obtained according to the corresponding recognition method. Therefore, it is hard to deal with the initial phase sensitivity of the complex HRRP in the test phase of the complex HRRP-based RATR.

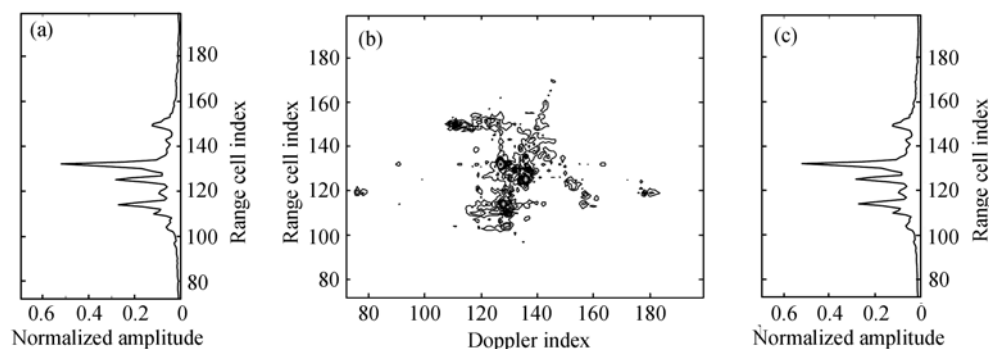


Figure 1 Explanation for the Parseval theorem using “Yark-42” measured data. (a) The 0.5-powered projection of the ISAR image onto the radial range dimensional; (b) the ISAR Image; (c) the 0.5-powered summation of all power profiles.

As discussed in section 2.1, because the average profile of a real HRRP frame can represent the stable SAT profile, the average profile of each frame can be used as the corresponding frame template in the real HRRP-based RATR. Then, can the average profile of a complex HRRP frame be used as the corresponding frame template in the complex HRRP-based RATR? As discussed in section 2.1, the ISAR image is the Fourier transformation of a complex HRRP frame along the range cell dimension, and the zero-Doppler slice in the ISAR image is identical with the average profile of the motion compensated complex HRRP frame, which is referred to as the average complex profile in this paper. The average real profile of a real HRRP frame can represent the projection of the target's scatterer distribution onto the radial range dimension, while the average complex profile of a complex HRRP frame only represents the zero-Doppler slice of the target's scatterer distribution. Therefore, such average complex profile cannot be used as the corresponding frame template for the complex HRRP-based RATR. Moreover, because the focus location in an ISAR image varies with the autofocus approach^[11], the average complex profile is not exclusive. This further demonstrates that the average complex profile is not the suitable frame template for RATR using the complex HRRP.

Therefore, the recognition algorithms and frame-template establishment methods used in the real HRRP-based RATR cannot be directly applied to the complex HRRP-based RATR.

3 A novel feature extraction method using the difference phase information in complex HRRP

3.1 The fundamental ideas

According to eq. (7), for the complex echoes from all range cells in a complex HRRP, their initial phases are the same. The assumed conditions are the same as those in section 2, and if the com-

plex echo from the n_1 th range cell is multiplied by the conjugate complex echo from the n_2 th ($n_1 \neq n_2$) range cell, then

$$y_{n_1 n_2}(m) = x_{n_1}(m) \cdot x_{n_2}^*(m) = \left[e^{j\psi(m)} \sum_{i_1=1}^{L_{n_1}} \sigma_{n_1 i_1} e^{j\phi_{n_1 i_1}(m)} \right] \cdot \left[e^{-j\psi(m)} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_2 i_2} e^{j\phi_{n_2 i_2}(m)} \right] \\ = \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \exp \left\{ -j \left\{ \frac{4\pi}{\lambda} [\Delta r_{n_1 i_1}(m) - \Delta r_{n_2 i_2}(m)] - [\psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}] \right\} \right\}. \quad (8)$$

Obviously, the product term $y_{n_1 n_2}(m)$ is independent of the initial phase. Let $L_{n_1 n_2} = L_{n_1} \cdot L_{n_2}$, $\sigma_{n_1 n_2 i} = \sigma_{n_1 i_1} \cdot \sigma_{n_2 i_2}$, $\Delta r_{n_1 n_2 i}(m) = \Delta r_{n_1 i_1}(m) - \Delta r_{n_2 i_2}(m)$, $\psi_{n_1 n_2 i 0} = \psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}$. Then, eq. (8) can be also written as

$$y_{n_1 n_2}(m) = \sum_{i=1}^{L_{n_1 n_2}} \sigma_{n_1 n_2 i} \exp \left\{ -j \left[\frac{4\pi}{\lambda} \Delta r_{n_1 n_2 i}(m) - \psi_{n_1 n_2 i 0} \right] \right\} = \sum_{i=1}^{L_{n_1 n_2}} \sigma_{n_1 n_2 i} e^{j\phi_{n_1 n_2 i}(m)}. \quad (9)$$

Let $n_2 - n_1 = J$. Compared with eq. (2), the product term $y_{n_1 n_2}(m)$ can also be regarded as the complex echo from a suppositional range cell in a special kind of complex HRRP, which is the vector product of the complex HRRP with J ($n_2 - n_1 = J$) range cells' delay and the conjugate vector of the original complex HRRP. In eq. (9), $L_{n_1 n_2}$ is regarded as the number of scatterers, $\sigma_{n_1 n_2 i}$ as the scatterer strength, $\Delta r_{n_1 n_2 i}(m)$ as the scatterer's radial distance, and $\psi_{n_1 n_2 i 0}$ as the initial phase of the i th scatterer. According to ref. [8], if x_{ni0} and y_{ni0} represent the initial cross and radial coordinates of the i th scatterer in the n th range cell, then the scatterer's radial displacement in the m th returned echo is

$$\Delta r_{ni}(m) = y_{ni0} [\cos(m \cdot \Delta\phi) - 1] + x_{ni0} \sin(m \cdot \Delta\phi), \quad (10)$$

where $\Delta\phi$ denotes the target-aspect difference between two neighboring echoes. Substituting eq. (10) into eq. (8), we obtain

$$y_{n_1 n_2}(m) = \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \exp \left\{ -j \left\{ \frac{4\pi}{\lambda} \left[(y_{n_1 i_1 0} - y_{n_2 i_2 0}) \cdot [\cos(m \cdot \Delta\phi) - 1] \right] + (x_{n_1 i_1 0} - x_{n_2 i_2 0}) \sin(m \cdot \Delta\phi) \right\} - [\psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}] \right\} \right\} \\ \approx \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \exp \left\{ -j \left\{ \frac{4\pi}{\lambda} \left[(y_{n_1} - y_{n_2}) \cdot [\cos(m \cdot \Delta\phi) - 1] \right] + (x_{n_1 i_1 0} - x_{n_2 i_2 0}) \sin(m \cdot \Delta\phi) \right\} - [\psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}] \right\} \right\} \\ = \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \{ \cos[\psi_{n_1 i_1 n_2 i_2 0} + \theta_{n_1 n_2}(m) + \phi_{n_1 i_1 n_2 i_2}(m) + j \sin[\psi_{n_1 i_1 n_2 i_2 0} + \theta_{n_1 n_2}(m) + \phi_{n_1 i_1 n_2 i_2}(m)]] \}, \quad (11)$$

where y_{n_1} and y_{n_2} represent the radial coordinates of the n_1 th and the n_2 th range cells, $\psi_{n_1 i_1 n_2 i_2 0} = \psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}$ denotes the scatterers' initial phase difference, $\theta_{n_1 n_2}(m) = -\frac{4\pi}{\lambda} (y_{n_1} - y_{n_2}) (\cos(m \cdot \Delta\phi) - 1)$ denotes the phase angle caused by the radial coordinates,

and $\varphi_{n_1 i_1 n_2 i_2}(m) = -\frac{4\pi}{\lambda}(x_{n_1 i_1 0} - x_{n_2 i_2 0})\sin(m \cdot \Delta\phi)$ denotes the phase angle caused by the cross coordinates. If $n_2 - n_1 = J$, the radar bandwidth $B=400$ MHz, the radar wavelength $\lambda \approx 5$ cm, and the HRRPs in about 3° target-aspect sector are taken as an HRRP frame, i.e., $(M-1)\Delta\phi \approx 3^\circ$ with the $M-1$ th returned echo vector being the last returned echo vector in an HRRP frame, then

$$\begin{aligned}\theta_{n_1 n_2}(M-1) &= -\frac{4\pi}{\lambda}(n_1 - n_2)\Delta y \{\cos[(M-1)\Delta\phi] - 1\} \\ &\approx \frac{4\pi}{0.05} \times J \times \frac{3.0 \times 10^8}{2 \times 400 \times 10^6} \times (\cos 3^\circ - 1) = -J \times 0.0411\pi,\end{aligned}\quad (12)$$

where $\Delta y = \frac{c}{2B}$ represents the radial coordinate difference between two neighboring range cells.

Therefore, when J is small, $|\theta_{n_1 n_2}(m)| \ll |\varphi_{n_1 i_1 n_2 i_2}(m)|$ and $\theta_{n_1 n_2}(m)$ can be ignored. Then,

$$\begin{aligned}y_{n_1 n_2}(m) &\approx \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \exp\{j[\psi_{n_1 i_1 n_2 i_2 0} + \varphi_{n_1 i_1 n_2 i_2}(m)]\} \\ &= \sum_{i_1=1}^{L_{n_1}} \sum_{i_2=1}^{L_{n_2}} \sigma_{n_1 i_1} \sigma_{n_2 i_2} \left\{ \cos[\psi_{n_1 i_1 n_2 i_2 0} + \varphi_{n_1 i_1 n_2 i_2}(m)] + j \sin[\psi_{n_1 i_1 n_2 i_2 0} + \varphi_{n_1 i_1 n_2 i_2}(m)] \right\}.\end{aligned}\quad (13)$$

The product term $y_{n_1 n_2}(m)$ in eq. (12) depends on the scatterers' initial phase difference, i.e., $\psi_{n_1 i_1 0} - \psi_{n_2 i_2 0}$, and the scatterers' cross coordinate differences, i.e., $x_{n_1 i_1} - x_{n_2 i_2}$ ($i_1 = 1, 2, \dots, L_{n_1}$ and $i_2 = 1, 2, \dots, L_{n_2}$). Then, similar to the power term of the real HRRP, i.e., eq. (4), $y_{n_1 n_2}(m)$ also consists of two terms, the auto-term and the cross-term, where the auto-term is the summation of the products of the strengths of the scatterers in the n_1 th and the n_2 th range cells and with approximately the same cross coordinates, and the cross-term is the summation of the products of the complex echoes from the scatterers in the n_1 th and the n_2 th range cells and with the different cross coordinates. Namely,

$$\begin{aligned}y_{n_1 n_2}(m) &\approx \sum_{i_1=1}^{L_{n_1}} \left\{ \sum_{i_{21}: x_{n_1 i_1} \approx x_{n_2 i_{21}}} \sigma_{n_1 i_1} \sigma_{n_2 i_{21}} \exp(j\psi_{n_1 i_1 n_2 i_{21} 0}) + \sum_{i_{22}: x_{n_1 i_1} \neq x_{n_2 i_{22}}} \sigma_{n_1 i_1} \sigma_{n_2 i_{22}} \exp\{j[\psi_{n_1 i_1 n_2 i_{22} 0} + \varphi_{n_1 i_1 n_2 i_{22}}(m)]\} \right\} \\ &= Y_{n_1 n_2} + \xi_{n_1 n_2}(m).\end{aligned}\quad (14)$$

Without scatterers' MTRC, the auto-term remains invariant, but the cross-term varies as a random variable with zero mean. Therefore, the average term of the corresponding product terms in a complex HRRP frame $\{y_{n_1 n_2}(0), y_{n_1 n_2}(1), \dots, y_{n_1 n_2}(M-1)\}$ can represent the stable auto-term approximately and reduce the cross-terms in the HRRP frame to a large degree. Moreover, the physical mechanisms of the auto-term and the cross-term are similar to those of the SAT and SCT in the real HRRP.

As the above analysis, if a complex HRRP with J range cells' delay (J is small) is multiplied

by the conjugate vector of the original complex HRRP, the physical mechanism of the product vector is similar to that of the real HRRP, and the difference phase information between the complex echoes from two range cells with J intervals remains. However, the amplitude property of the product vector obtained by such method is unsuitable for the recognition. Refs. [5,6] proposed the power transformation or the Box-Cox transformation as the preprocessing method for the real HRRP to improve the recognition performance of the real HRRP. It was shown that the transformed echoes from all range cells approximate to follow Gaussian distribution when the power parameter α is proper (usually $0 < \alpha < 1$). Therefore, in the classifier based on the Euclidean distance or the Gaussian distribution assumption, the optimal power transformation or Box-Cox transformation for the real HRRP can greatly improve the recognition performance. Refs. [9, 10] made a detailed analysis on the physical mechanism of the power transformation or the Box-Cox transformation on improving the real HRRP-based recognition performance. It was shown that such preprocessing method with proper power parameter α (usually $0 < \alpha < 1$) can reduce the amplitude's dynamic range of the real HRRP, increase the contribution of weak scatterers to the similarity measurement, and restrain the speckle effect of the real HRRP (which is unfavorable for the recognition). Therefore, the similarity measurement using Euclidean distance and the Gaussian distribution assumption will be more reasonable for the optimal power transformed or Box-Cox transformed real HRRP-based RATR. Although the viewpoints about the reason that the power transformation or the Box-Cox transformation improves the recognition performance in refs. [5,6,9,10] are different, their experimental results all showed that the power transformation or the Box-Cox transformation can improve recognition performance only when the power parameter $0 < \alpha < 1$; otherwise, the recognition performance will be degraded. The above feature extraction method uses the conjugate product between the complex echoes from two range cells with J intervals (J is small) to counteract the initial phase, yet the complex feature vector is in power form (which is similar to the power profile in section 2.1, but with the phase information in the complex HRRP), and because of the amplitudes of the two complex echoes being similar, the amplitude profile of the product vector can increase the amplitude's dynamic range of the original complex HRRP, reduce the contribution of weak scatterers to the similarity measurement, and enlarge the speckle effect. Under the limit condition $J = 0$, the product vector corresponds to the power transformed real HRRP with $\alpha = 2$, i.e., the power profile in section 2.1. According to what was analyzed above, this is unfavorable for the recognition from the viewpoint of using the HRRP's amplitude property. If the initial phase can be counteracted without increasing the amplitude's dynamic range of the original complex HRRP, not only the phase information can be applied to the recognition, but also the amplitude property of the complex feature vector is favorable for the recognition. There is a theorem for two Gaussian variables with zero means and same variances, which will be proved in the Appendix.

Theorem 1. Let X and Y be two independent random variables which follow the same Gaussian distribution with zero means and same variances, i.e., $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$. If

the random variable $V = \frac{XY}{\sqrt{X^2 + Y^2}}$, then $V \sim N\left(0, \frac{\sigma^2}{4}\right)$.

Namely, if the product term of the two Gaussian variables is divided by a conversion factor, where the conversion factor can convert the random variable in power form into the one in am-

plitude form, the quotient term as a random variable still follows Gaussian distribution. We applied this theorem to the feature extraction for the complex HRRP, and obtain the complex feature vector

$$z_J(m) = \left[z_{J1}(m), z_{J2}(m), \dots, z_{Jn}(m), \dots, z_{J(N-J)}(m) \right]^T$$

$$= \left[\frac{x_1(m) \cdot x_{1+J}^*(m)}{\sqrt{|x_1(m)|_2^2 + |x_{1+J}^*(m)|_2^2}}, \frac{x_2(m) \cdot x_{2+J}^*(m)}{\sqrt{|x_2(m)|_2^2 + |x_{2+J}^*(m)|_2^2}}, \dots, \frac{x_{N-J}(m) \cdot x_N^*(m)}{\sqrt{|x_{N-J}(m)|_2^2 + |x_N^*(m)|_2^2}} \right]^T. \quad (15)$$

The complex feature vector shown in eq. (15) has been converted into the one in amplitude form (which is similar to the original HRRP, while not similar to the power profile, and with the phase information in the complex HRRP). For both large echoes, their product term is divided by a large conversion factor, while for both small echoes, their product term is divided by a small conversion factor; therefore, such feature extraction method can reduce the amplitude's dynamic range of the product vector and the amplitude property of the complex feature vector is more suitable for recognition. The complex feature vector in eq. (15) is referred to as the complex feature vector with difference phases in this paper. According to what was discussed above, such complex feature vector can be regarded as a complex echo vector consisting of the discrete echo components from the suppositional range cells. In addition, according to the scattering center model, there are three kinds of scatterer distribution in a range cell^[8], of which the complex echoes from the first and second types of cells both follow Gaussian distribution and the proportion of the first and second types of cells in the three types of cells is large. Nevertheless, the mean of the complex echo from the second type of cell is nonzero, and the variances of the complex echoes from two range cells are different. Since J is small, the variances of the complex echoes from the two range cells with J intervals approximate to be the same. Moreover, the above theorem can be generalized to the random variable with nonzero mean. Thus, the components in the complex feature vector with difference phases approximate to follow Gaussian distribution, accordingly by which the statistical recognition is simple.

3.2 Two problems in applications

The fundamental idea of the feature extraction method using the difference phase information in the complex HRRP has been proposed in section 3.1, but there are still two problems discussed.

Firstly, according to eq. (15), there is a parameter, cell interval parameter J , in the complex feature vector with difference phases; thereby how to select the cell interval parameter J is to be discussed. In an ideal case, if the returned echoes are sampled critically, namely the sampling frequency is equal to the radar bandwidth; the echoes from the neighboring range cells in an HRRP are independent. Actually, because the target returned echoes are usually multiplied by window functions before getting the HRRP data (e.g., the Hamming window was applied to obtain the measured HRRP data in this paper.) in order to be denoised, the range resolution decreases and the echoes from the neighboring range cells in an HRRP are not completely independent, which does not satisfy the theorem introduced in section 3.1. Moreover, the feature vector with $J=1$ is similar to the power profile of the real HRRP. According to section 3.1 and refs. [5,6,9,10], the recognition performance of such feature vector is not good. Therefore, the cell interval parameter should be larger than one. However, if J is too large, the complex echoes from

different parts of a target and with distinct amplitudes are used in the feature extraction method. Then, the product operation results in the echoes from strong scatterers decreasing and the echoes from weak scatterers increasing, and the components in the signal support region of the complex feature vector approximate to have the same amplitudes. Thereby, the target discriminant information for the recognition decreases. Figure 2 shows the normalized amplitude profile of a complex HRRP in the measured “Yark-42” data and the normalized amplitude profiles of its complex feature vectors with $J=1, 2, 10$. Obviously, compared with the amplitude profile of the complex HRRP shown in Figures 2(a) and 2(c) which show the amplitude profiles of the original sample and the complex feature vector with $J=2$, the amplitude’s dynamic range increases and some echoes of weak scatterers further decrease in Figure 2(b) which shows the amplitude profile of the complex feature vector with $J=1$; while all of the components in the signal support region approximate to have the same amplitudes without any target discriminant information except for the length of the signal support region in Figure 1(d) which shows the amplitude profile of the complex feature vector with $J=10$. Therefore, the cell interval parameter J should be larger than one but not too large. The recognition performances of the complex feature vectors with different cell interval parameters will be compared based on the measured data in section 4.

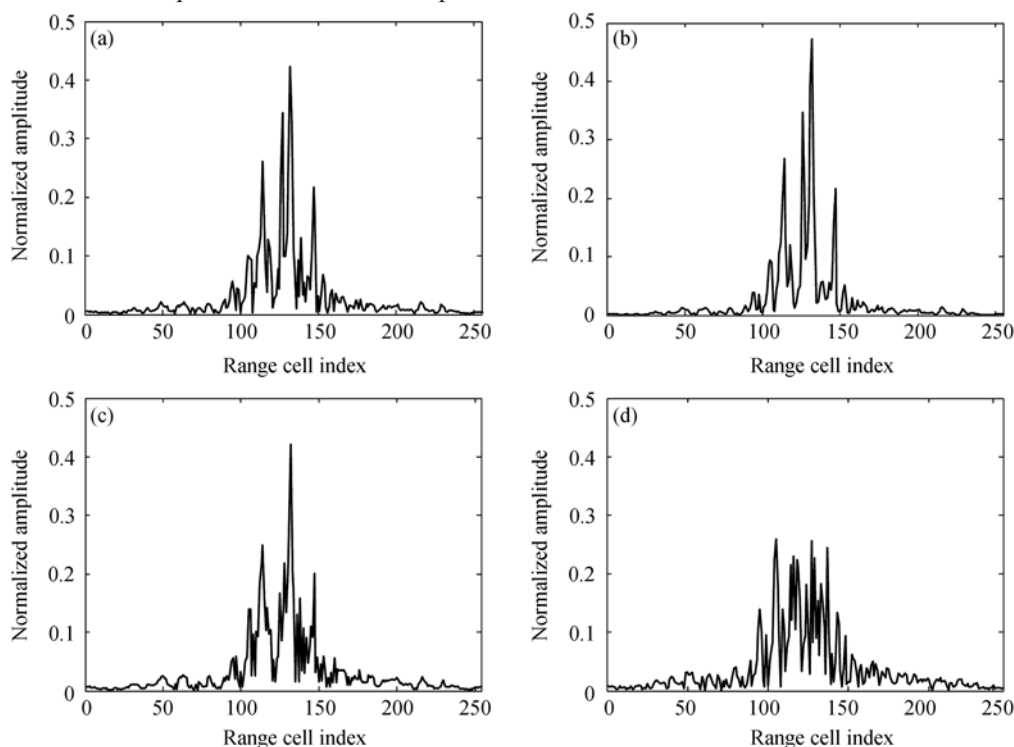


Figure 2 The normalized amplitude profiles of a complex HRRP and its complex feature vectors with difference phases in “Yark-42” measured data. (a) The original complex HRRP; (b) the complex feature vector with $J=1$; (c) the complex feature vector with $J=2$; (d) the complex feature vector with $J=10$.

Secondly, according to the ideas in section 3.1, the complex feature vector with difference phases is the product vector divided by the corresponding conversion factors; therefore, the range cell order of a complex HRRP will influence the feature extraction. However, due to the time-shift sensitivity, the range cell order of an HRRP varies with the range window, and all of

the samples used in the training and test phases should be time-shift compensated^[7, 8]. Thus, the complex feature vector with difference phases extracted from an original complex HRRP without time-shift compensation should also be time-shift compensated in the recognition. Since the components in a complex feature vector with difference phases are the combination term of the complex echoes from the two range cells with J intervals in an original complex HRRP, the complex echoes from two range cells are involved in the time-shift compensation for such complex feature vector, which will reduce the time-shift compensation precision. Refs. [8, 12] showed that the slide correlation processing is a simple and effective time-shift compensation method for the HRRP. If all of the complex HRRPs used in the training and test phases are time-shift compensated by the slide correlation processing, then the complex feature vectors with difference phases extracted from the time-shift compensated complex HRRPs should be independent of the time shift and can be used to classifiers directly. Figure 3 takes the minimum Euclidean distance classifier as an example to show how to apply the feature extraction method proposed in this paper to classifiers. According to Figure 3, since the complex feature vector with difference phases should be extracted from a time-shift compensated complex HRRP in the test phase, the average real profiles of all frames should be stored in the template database for the time-shift compensation in the test phase. The complex feature vector with difference phases extracted from a time-shift compensated complex HRRP is independent of the time shift and can be directly used to calculate the Euclidean distances between the complex feature templates with difference phases of all frames without searching the minimum time-shift compensated Euclidean distance, which is different from the process of applying the real HRRP to the minimum Euclidean distance classifier.

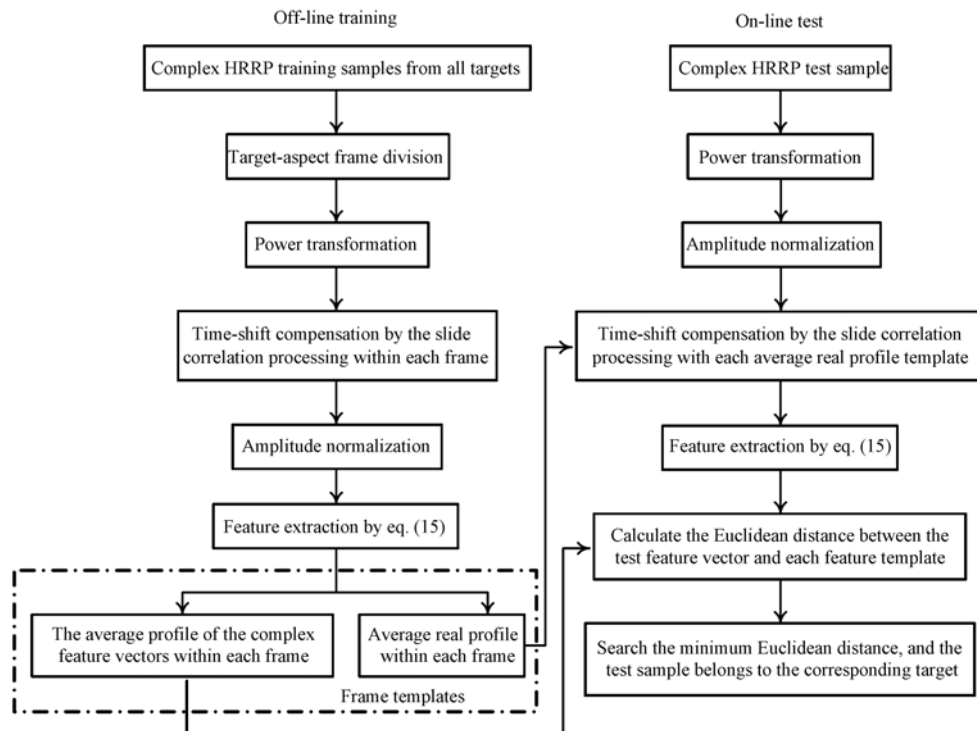


Figure 3 The flow chart of the minimum Euclidean distance classifier using the proposed feature extraction method.

4 Experimental results

4.1 Introduction to measured data

The results presented here are based on measured airplane data. The parameters of the targets and radar are shown in Table 1, and the projections of target trajectories onto ground plane are shown in Figure 4, from which the aspect angle of the airplane can be estimated according to its relative position to radar. As shown in Figure 4, the measured data are segmented. Training data and test data are from different data segments, among which the 2nd and the 5th segments of Yark-42, the 5th and the 6th segments of An-26, and the 6th and the 7th segments of Cessna Citation S/ II are taken as the training data, other data segments are taken as the test data. The training data cover almost all of the target-aspect angles, but their elevation angles are different from those of the test data. With large SNR, the effect of noise in the measured data on the recognition can be ignored. The measured complex HRRP is a 256-dimensional complex vector.

Table 1 The parameters of planes and radar in the ISAR experiment

Planes	Radar parameters	Center frequency: 5520 MHz; Bandwidth: 400 MHz	
	Length (m)	Width (m)	Height (m)
Yark-42	36.38	34.88	9.83
An-26	23.80	29.20	9.83
Cessna Citation S/II	14.40	15.90	4.57

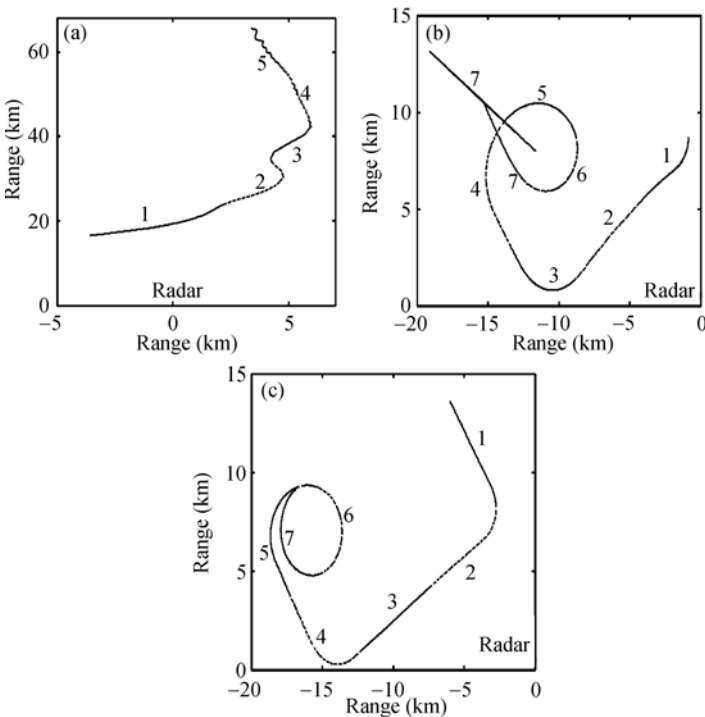


Figure 4 The projections of target trajectories onto ground plane. (a) Yark-42; (b) An-26; (c) Cessna Citation S/II.

4.2 Recognition results

Figures 5(a) and 5(b) show the recognition rates of the proposed complex feature vectors with $J = 1, 2, 3, 4, 5$ and the real HRRP versus the power parameter α in the minimum Euclidean distance classifier^[7-9] and the AGC^[6,8]. From the recognition results, we know that: a) for both of

the classification algorithms, the optimal cell interval parameter $J^*=2$, and the recognition rates of the complex feature vectors with $J=2,3,4$ are larger than those of the real HRRP; b) the largest recognition rates of the complex feature vector with difference phases (when $\alpha^* = 0.1$, $J^* = 2$ in the minimum Euclidean classifier, and $\alpha^* = 0.3$, $J^* = 2$ in the AGC) are about 5 percentage points larger than those of the real HRRP (when $\alpha^* = 0.1$ in the minimum Euclidean classifier and $\alpha^* = 0.5$ in the AGC); c) without the power transformation, i.e., power parameter $\alpha = 1$, the recognition rates of the complex feature vector with $J^* = 2$ in the minimum Euclidean distance classifier and the AGC are larger than those of the real HRRP by approximately 12 and 5 percentage points, respectively; d) the variations of the recognition rates of the minimum Euclidean distance classifier and the AGC with the power parameter are different, and mostly the AGC has the better recognition performances. The reason is that the statistical assumptions in the two methods are different^[8]. The minimum Euclidean distance classifier only utilizes the mean parameter of an HRRP frame, while the AGC utilizes not only the mean parameter, which can reflect the scatterer strength information in each range cell, but also the variance parameter, which can reflect the scatterer distribution information in each range cell. Therefore, the proposed feature extraction method can effectively use the difference phase information in the complex HRRP and improve the recognition performance if only the cell interval parameter is properly selected.

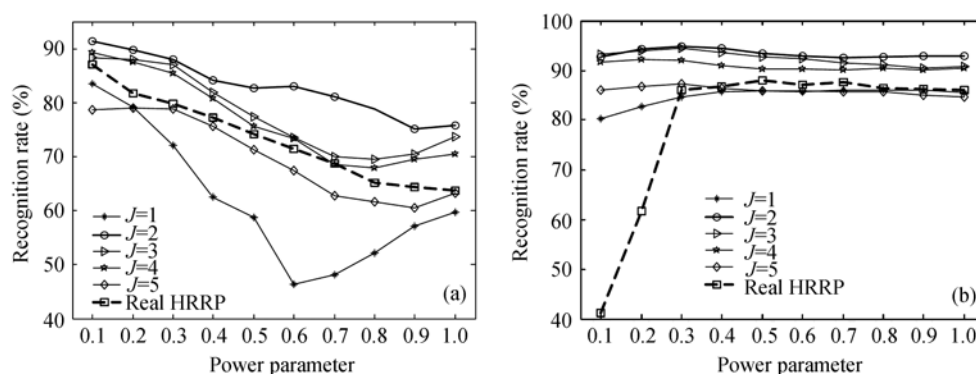


Figure 5 The average recognition rates of the real HRRP and the complex feature vectors with $J=1,2,3,4,5$ versus the power parameter in the minimum Euclidean distance classifier and the AGC. (a) The minimum Euclidean distance classifier; (b) the AGC.

5 Conclusion

Most literatures only used the real HRRP, i.e., the amplitude information in the complex HRRP, for RATR, whereas the phase information was discarded. This paper points out the fact that the initial phase sensitivity restricts the application of the complex HRRP to RATR. Then, a novel feature extraction method which can counteract the initial phase and reserve the difference phase information between range cells in a complex HRRP is proposed. The recognition algorithms, frame-template establishment methods and preprocessing methods used in the real HRRP-based RATR can also be applied to the proposed complex feature vector-based RATR. Moreover, the components of the complex feature vector with difference phases approximate to follow Gaussian distribution, which makes it simple to perform the statistical recognition by such complex feature

vector. The recognition experiments based on measured data show that the complex feature vector with difference phases can obtain better recognition results than the real HRRP in the minimum Euclidean distance classifier and the AGC if only the cell interval parameters are properly selected. Therefore, the proposed feature extraction method can effectively use the difference phase information in the complex HRRP and improve the recognition performance. The future work is about proposing similar methods for other kinds of radar target recognition, for example the application discussed in ref. [13].

Appendix

Let X and Y be two independent random variables which follow the same Gaussian distribution with zero means and the same variances, i.e., $X \sim N(0, \sigma^2)$ and $Y \sim N(0, \sigma^2)$. If the random variables $R = \sqrt{X^2 + Y^2}$ and $\Theta = \arctg\left(\frac{Y}{X}\right)$, then the joint probability density function (PDF) of R and Θ is

$$f_{R,\Theta}(r, \theta) = \frac{r}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right). \quad (\text{A1})$$

If the random variables $U = \frac{X^2 - Y^2}{\sqrt{X^2 + Y^2}}$ and $V = \frac{XY}{\sqrt{X^2 + Y^2}}$, then $R = \sqrt{U^2 + 4V^2}$, $\Theta = \frac{1}{2} \arctg\left(\frac{2V}{U}\right)$. Now, let

$$J(u, v) = \begin{vmatrix} \frac{\partial r}{\partial u} & \frac{\partial r}{\partial v} \\ \frac{\partial \theta}{\partial u} & \frac{\partial \theta}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{u}{\sqrt{u^2 + 4v^2}} & \frac{4v}{\sqrt{u^2 + 4v^2}} \\ \frac{-v}{u^2 + 4v^2} & \frac{u}{u^2 + 4v^2} \end{vmatrix} = \frac{u^2}{(u^2 + 4v^2)^{\frac{3}{2}}} + \frac{4v^2}{(u^2 + 4v^2)^{\frac{3}{2}}} = \frac{1}{\sqrt{u^2 + 4v^2}}. \quad (\text{A2})$$

The joint PDF of U and V then becomes

$$f_{U,V}(u, v) = |J(u, v)| f_{R,\Theta}\left(\sqrt{u^2 + 4v^2}, \frac{1}{2} \arctg\left(\frac{2v}{u}\right)\right) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + 4v^2}{2\sigma^2}\right), \quad (\text{A3})$$

and V follows the distribution of

$$f_V(v) = \int_{-\infty}^{+\infty} f_{U,V}(u, v) du = \int_{-\infty}^{+\infty} \frac{1}{2\pi\sigma^2} \exp\left(-\frac{u^2 + 4v^2}{2\sigma^2}\right) du = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{2v^2}{\sigma^2}\right), \quad (\text{A4})$$

namely, $V \sim N\left(0, \frac{\sigma^2}{4}\right)$.

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